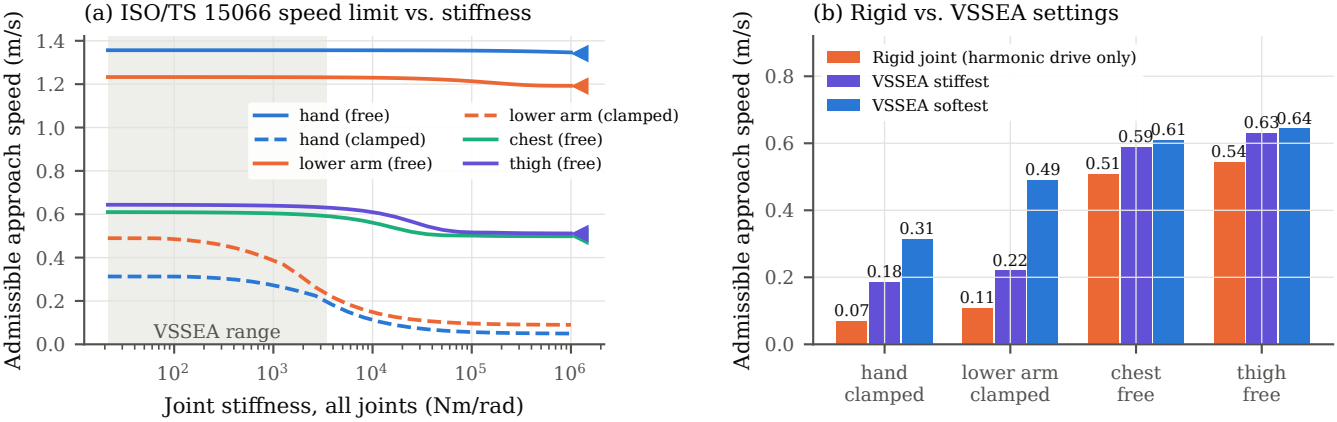
(a) ISO/TS 15066 speed limit vs. stiffness
Admissible approach speed (m/s)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
hand (free)
hand (clamped)
lower arm (free)
lower arm (clamped)
chest (free)
thigh (free)
VSSEA range
$10^2$
$10^3$
$10^4$
$10^5$
$10^6$
Joint stiffness, all joints (Nm/rad)
(b) Rigid vs. VSSEA settings
Admissible approach speed (m/s)
0.8
0.6
0.4
0.2
0.0
Rigid joint (harmonic drive only)
VSSEA stiffest
VSSEA softest
0.07
0.18
0.31
0.11
0.22
0.49
0.51
0.59
0.61
0.54
0.63
0.64
hand clamped
lower arm clamped
chest free
thigh free